VICTORY 2027

SOUND

LECTURE-03 PHYSICS

Shailendra sir



Topics To Be Covered

1. REVISION
2. VIBRATIONS
3. RESONANCE
4. QUESTIONS


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Revision

Reflection

$v_s = \frac{2d}{t}$ Echo

Vibration



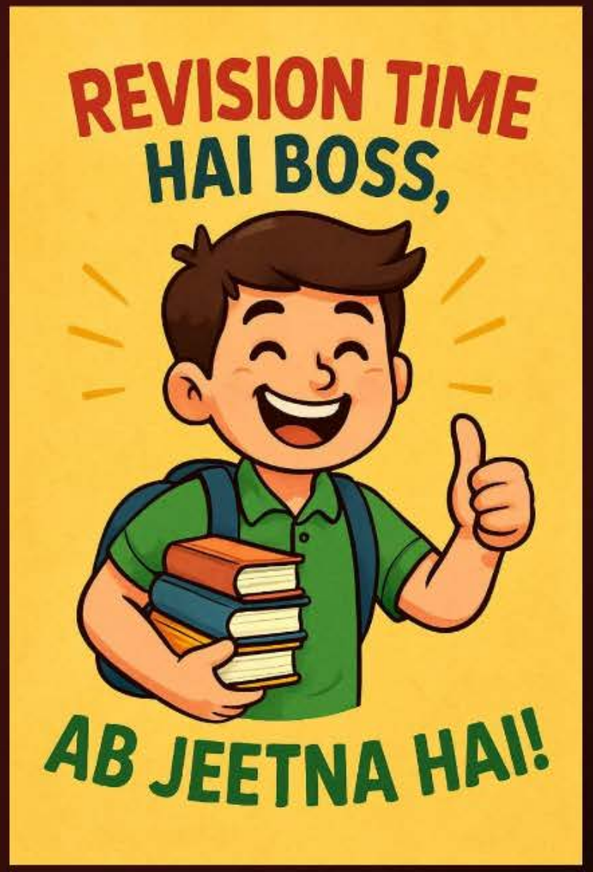
Mean Position

$F = 0$
$\tau = 0$

① Natural 

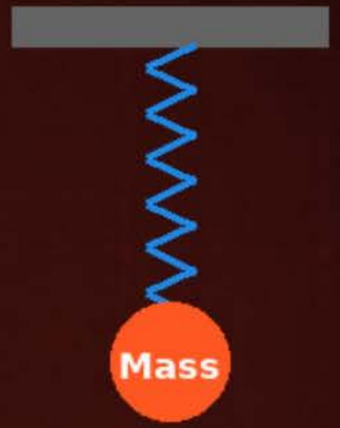
② Damped 

③ Forced 

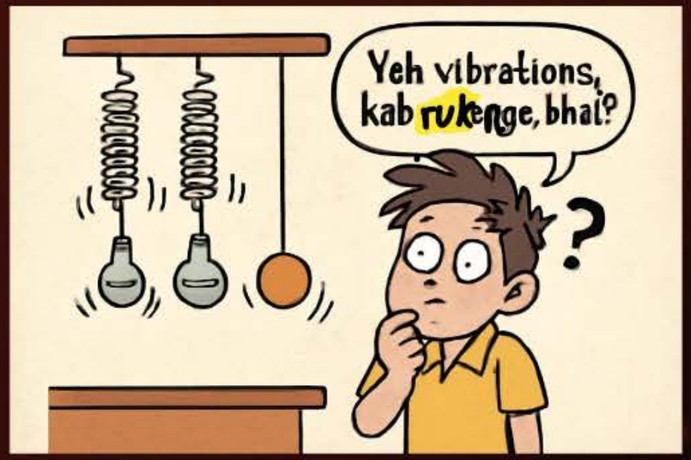


NATURAL VIBRATIONS

A body clamped at one point when disturbed slightly from its rest position, starts vibrating. The vibrations so produced are called the **natural or free vibrations** of the body. **The periodic vibrations of a body in the absence of any external force on it, are called the natural (or free) vibrations.** The time period of the body is called its **natural period** and the frequency is called its **natural frequency**.



- The period (or frequency) of vibration depends on the shape and size (or structure) of the body.
- Each body, capable of vibrating, vibrates with a constant natural frequency and its amplitude of vibration remains constant.



Examples Of Natural (Or Free) Vibrations

(1) If the bob of a simple pendulum is displaced slightly from its mean (or rest) position, it starts vibrating with its natural frequency

$$f = \frac{1}{2\pi}\sqrt{\frac{g}{l}} \Rightarrow T = 2\pi\sqrt{\frac{l}{g}}$$

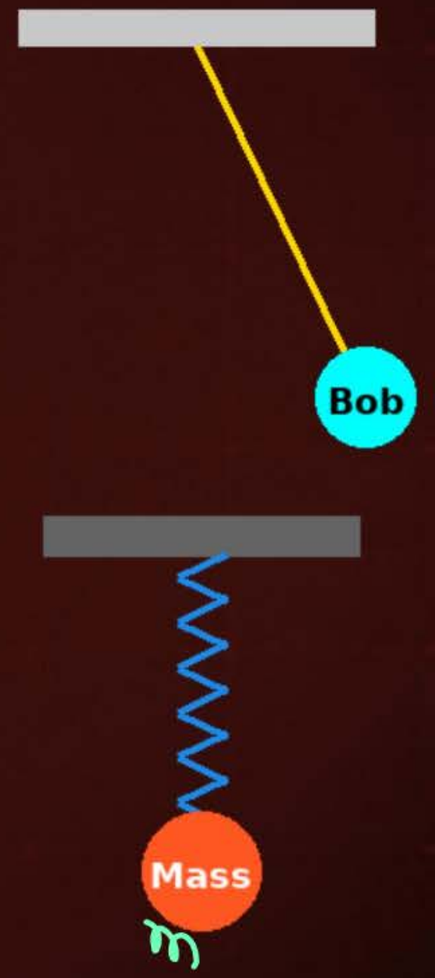
A simple pendulum of length 1.0 m on earth surface, where the acceleration due to gravity is $9.8\ \mathrm{ms}^{-2}$, has its natural frequency 0.5 Hz.

(2) A load suspended from a spring, when stretched (or compressed) and then released, starts vibrating with its natural frequency.

$$f = \frac{1}{2\pi}\sqrt{\frac{K}{m}} \Rightarrow T = 2\pi\sqrt{\frac{m}{K}}$$

K = Spring Constant

(3) When a tuning fork is struck against a hard rubber pad, it vibrates with its natural frequency. The vibrations form longitudinal waves in air, which when reach our ears, sound is heard. This **sound is of a single frequency** (i.e., it is a **pure note**).



Examples Of Natural (Or Free) Vibrations

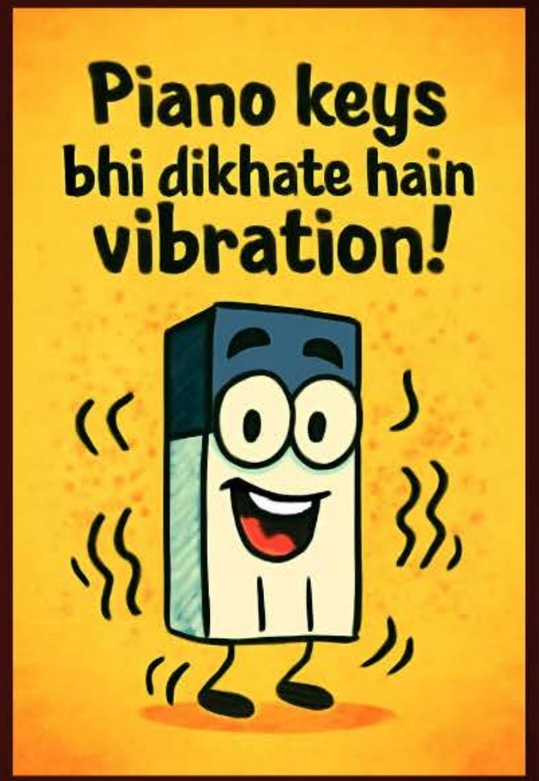
(4) When we strike the keys of a piano, various strings are set in vibration each at its natural frequency.

(5) When an **air column in a flute (or organ pipe)** is made to vibrate, it vibrates with its natural frequency which is inversely proportional to the length of air column. In a flute, the notes of different frequencies are produced by changing the effective length of air column when different holes in it are closed.

- In an organ pipe open at both ends, different modes of vibrations are of frequencies in the ratio **1 : 2 : 3.....** while in an organ pipe with one end closed, the frequencies of different modes are in ratio **1 : 3 : 5 : .....**

$f_1 : f_2 : f_3 \text{---} = 1 : 2 : 3 \text{---}$

↳ odd harmonics



Examples Of Natural (Or Free) Vibrations

(6) When the string in the instruments like **sitar, guitar, violin**, etc. is once plucked, the transverse vibrations of a definite natural frequency are produced in the string. Frequency of note produced by a string stretched between its ends is given as

$$f = \frac{1}{2l}\sqrt{\frac{T}{\pi r^2 d}}$$

T → Tension

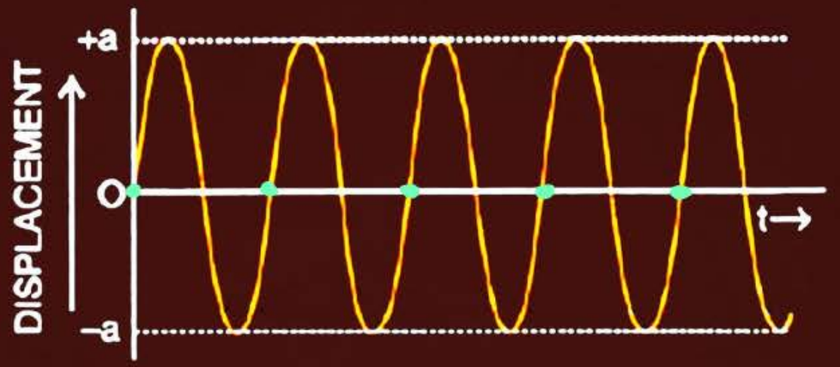
d → density

Where d is the density of material of the string and $\pi r^2 d = m$ is the mass of unit length of the string.



Nature Of Natural Vibrations

The natural vibrations are the **simple harmonic vibrations** under the restoring force, the amplitude and frequency of which remain constant. Once a body starts vibrating, it continues its vibrations with the same amplitude and same frequency forever.



The vibrations of a constant amplitude can occur only in vacuum. In practice, the surrounding medium offers resistance (or friction) to the motion, so the energy of vibrating body continuously decreases due to which the amplitude of motion gradually decreases.

Natural vibrations ka nature:



kabhi rukta nahi!

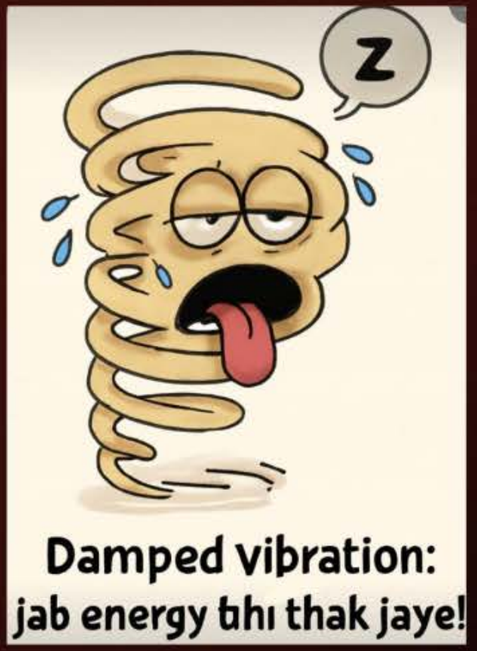
DAMPED VIBRATIONS

It is our common experience that when a body is made to vibrate in a medium, the amplitude of the vibrating body continuously decreases with time and ultimately the body stops vibrating. Such vibrations are called the damped vibrations.

The periodic vibrations of a body of decreasing amplitude in presence of a resistive force are called the damped vibrations.

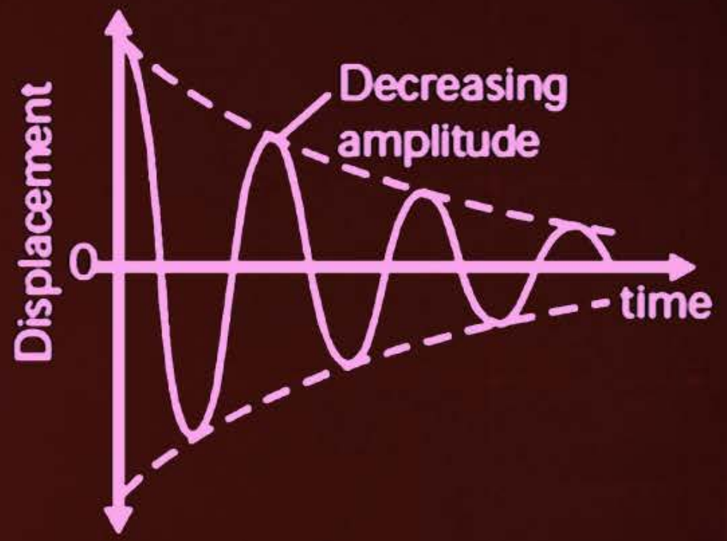
In damped vibrations, two forces act on the body : (i) the restoring force, and (ii) frictional (or resistive) force due to the surrounding medium. The amplitude of motion decreases due to the frictional force.





DAMPED VIBRATIONS

- The **frictional force (damping force) at any instant is proportional to the velocity of the body** and it has the tendency to resist the motion.
- As a result, the energy of the vibrating body continuously dissipates in doing work against the force of friction and so its amplitude gradually decreases. After some time, when it has lost all its energy, it stops vibrating.
- The energy lost by the vibrating body changes continuously into the heat energy and it gets dissipated in the surrounding medium.
- The rate at which the energy is lost to the surroundings (or the rate of decrease of amplitude) depends on the nature (i.e., viscosity, density, etc.) of the surrounding medium and also on the shape and size of the vibrating body.



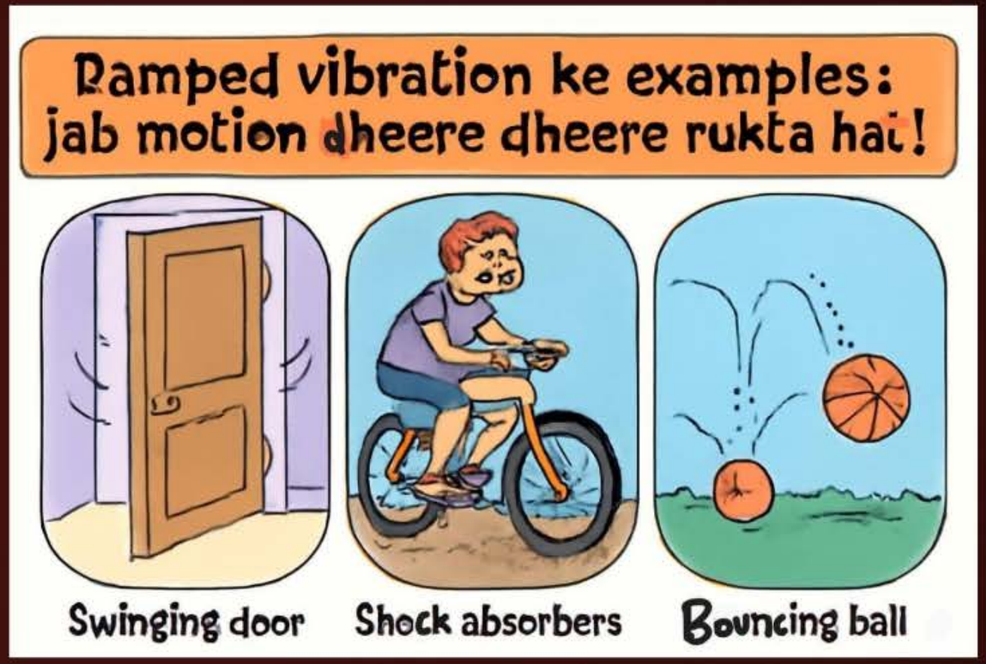
EXAMPLES OF DAMPED VIBRATIONS

(1) When a slim branch of a tree is pulled and then released, it makes the damped vibrations.

(2) A tuning fork when stroked on a rubber pad, executes the damped vibrations in air.

(3) A simple pendulum oscillating in air (or any other medium) executes the damped vibrations.

(4) The vibrations of a loaded spring in air are the damped vibrations.



Difference Between The Natural And Damped Vibrations

HW

	Free vibrations	Damped vibrations
1.	The amplitude of free vibrations remain constant and the vibration continue forever.	The amplitude of damped vibrations gradually decreases with time and ultimately the vibrations cease.
2.	There is no loss of energy in free vibrations.	In each vibration, there is some loss of energy in the form of heat.
3.	No external force acts on the vibrating body. The vibrations are only under the restoring force.	The frictional or damping force acts to oppose the motion.
4.	The frequency of vibrations remains constant equal to the natural frequency.	The frequency of vibrations is less than the natural frequency


Free vibration:
mast chal, raha hai?
Damped vibratiom:
thak gaya, rok jouur?

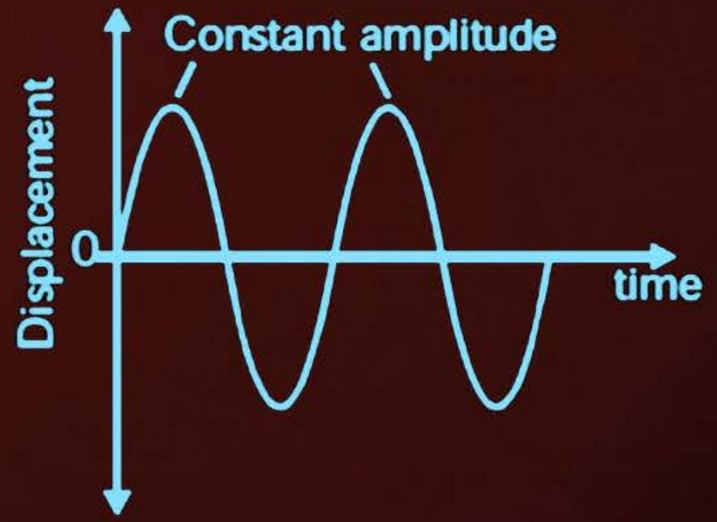
FORCED VIBRATIONS

The amplitude of natural vibrations of a body vibrating in a medium cannot remain constant due to the presence of damping forces of the surrounding medium. However, it is possible to keep the amplitude of vibrations constant by applying an external periodic force such that the external periodic force compensates for the loss of energy in each vibration due to the damping forces. The vibrations of the body then are called the **forced vibrations**.

The vibrations of a body which take place under the influence of an external periodic force acting on it, are called the forced vibrations.

The body executing the forced vibrations is acted upon by three forces:

(i) the restoring force,
(ii) the frictional (or resistive) force, and
(iii) the external periodic force.





FORCED VIBRATIONS

- When the external periodic force is applied on the vibrating body, the body no longer vibrates with its own natural frequency, but it gradually acquires the frequency of the applied periodic force. The external applied force is also called the **driving force**.
- The amplitude of the forced vibrations depends on the frequency of the external force.
- If the frequency of the external force is different from the natural frequency of the body, the amplitude of oscillations is very small.
- But if the frequency of the external force is exactly equal to the natural frequency of the vibrating body, the amplitude of oscillations is very large.
- However the amplitude of forced vibration does not change with time.


Forced vibration: boss,
tum hi to dilo ko hilate ho!



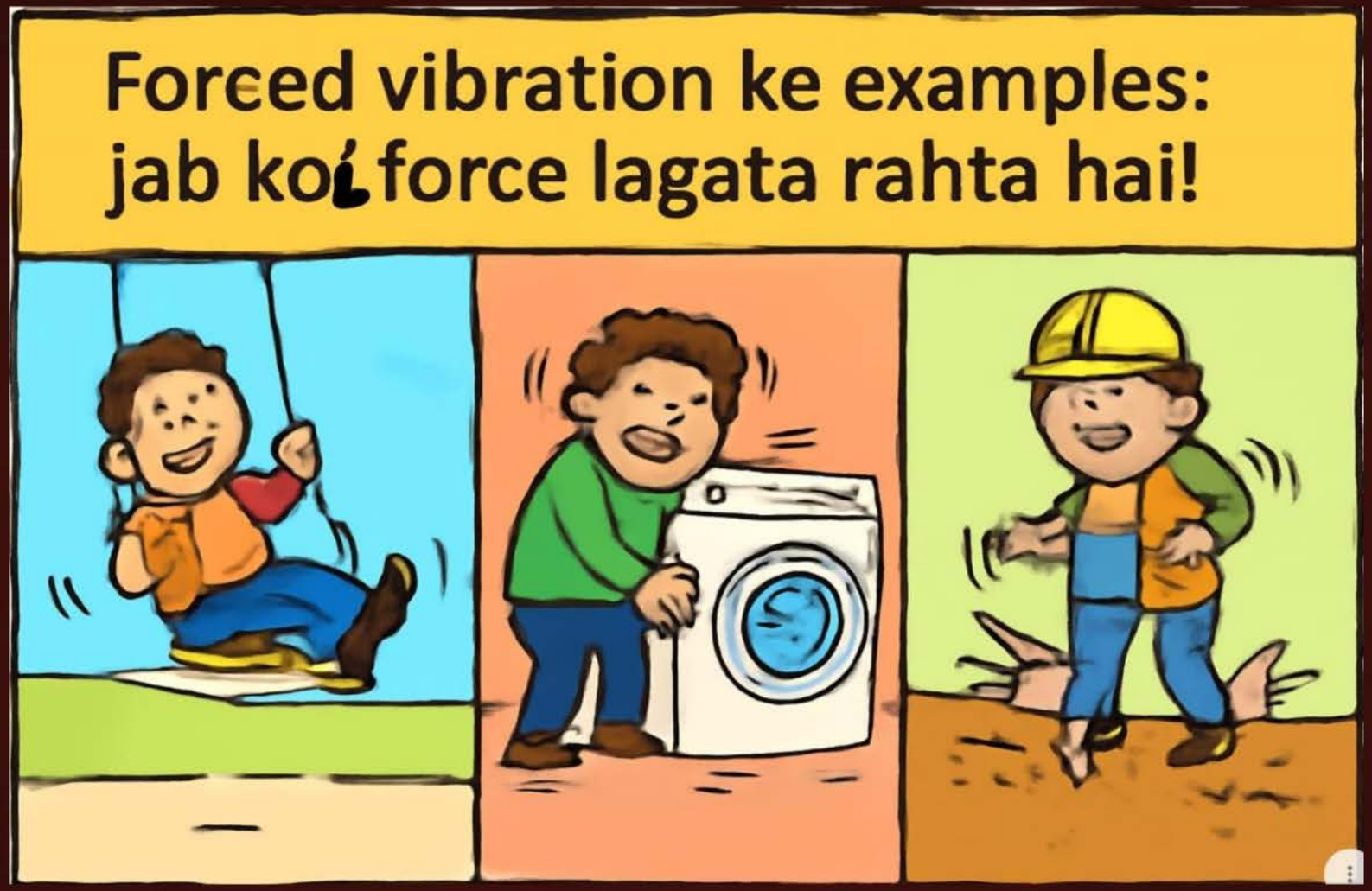
Examples Of Forced Vibrations

(1) When a vibrating tuning fork is pressed against the top of a table, the tuning fork forces the table top to vibrate with its own frequency. The vibrations produced in the table top are the forced vibrations. The table top has a much larger vibrating area than the tuning fork, so the forced vibrations of the table top send forth a greater energy and produces a louder (or more intense) sound than that produced by the tuning fork.

(2) The vibrations produced in the diaphragm of a microphone sound box with frequencies corresponding to the speech of the speaker, are the forced vibrations.

(3) In a guitar, an artist applies the periodic force on the strings of the guitar to execute the forced vibrations in them.

(4) In stringed instruments provided with a hollow sound box containing air, strings are made to vibrate by plucking which produces forced vibrations in air of the sound box. The surface area of air in the sound box is large, so the forced vibrations of air send forth a greater energy and cause a loud sound.


Forced vibration ke examples:
jab koi force lagata rahta hai!



QUESTIONS

Free vibrations are:

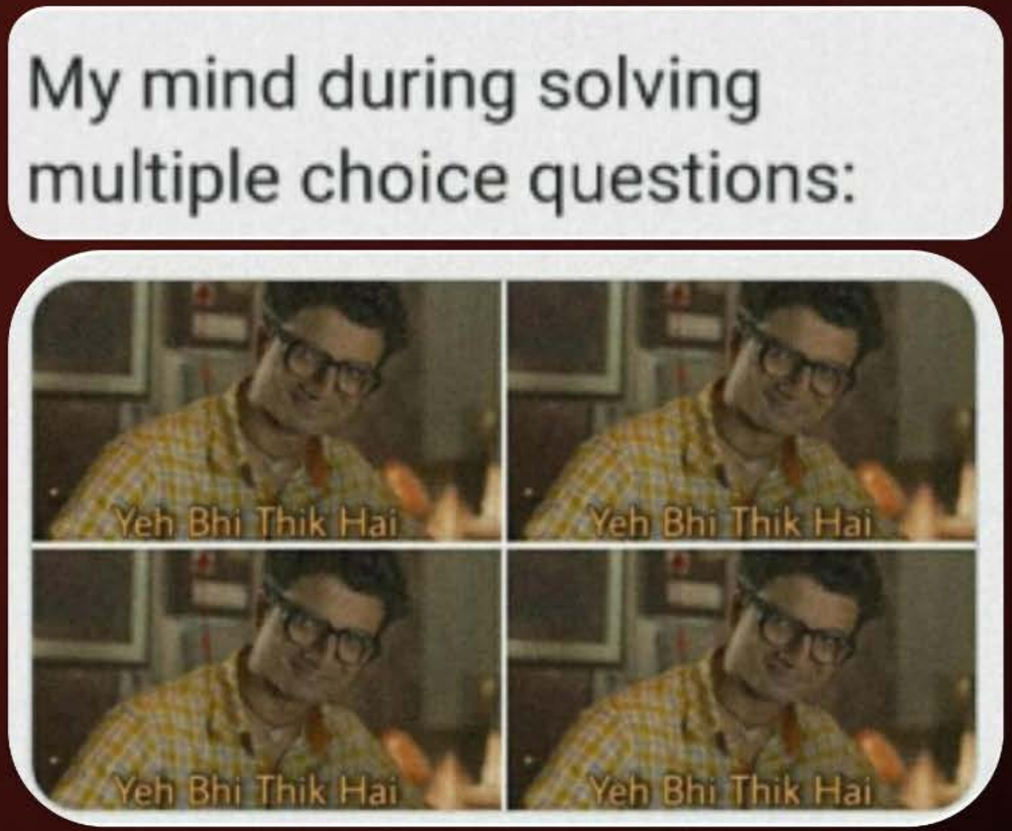
1. The vibrations under the influence of a periodic force
2. The vibrations with larger amplitude
3. The vibrations when the frequency continuously decreases
4. The vibrations with a constant frequency and constant amplitude.



QUESTIONS

The vibrations of a body that take place under the influence of an external periodic force are called:

1. natural vibrations
2. forced vibrations
3. free vibrations
4. damped vibrations



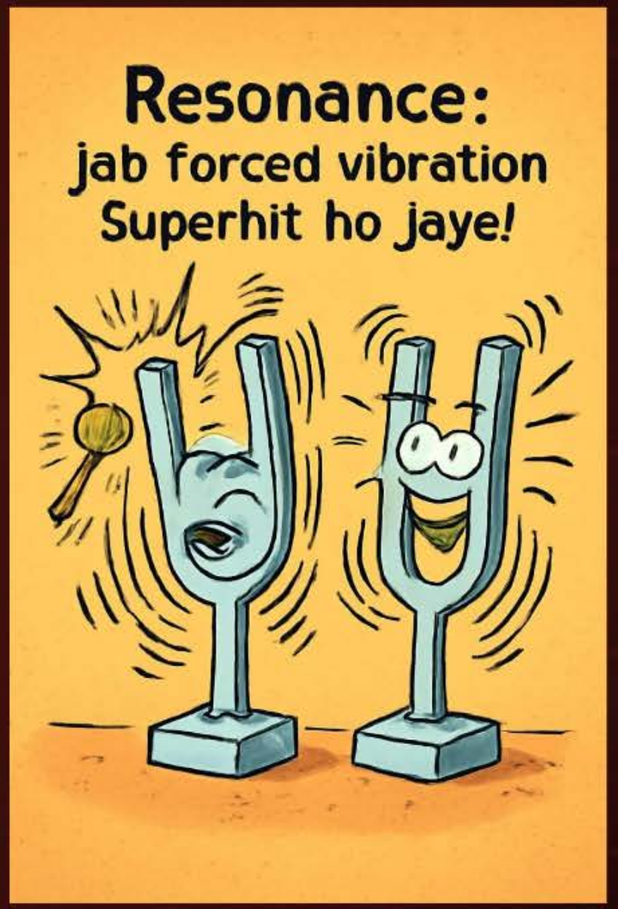
RESONANCE (A Special Case Of Forced Vibrations)

Resonance is a special case of forced vibrations. **When the frequency of the externally applied periodic force on a body is equal to its natural frequency**, the body readily begins to **vibrate with an increased amplitude**. This phenomenon is known as resonance. The vibrations of large amplitude are called the resonant vibrations.

Condition for Resonance:

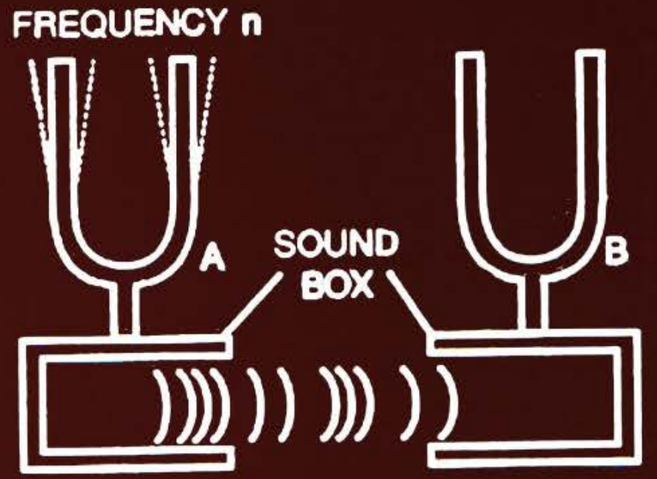
Resonance occurs only when the frequency of the applied force is exactly equal to the natural frequency of the vibrating body and the applied force causes forced vibration in the body.

At resonance, a **loud sound is heard** because the body vibrating with a very large amplitude sends forth a large amount of energy in the medium.



Resonance with Tuning Forks

Mount two identical tuning forks A and B of the same frequency on two separate sound boxes such that their open ends face each other. **If the prong of one of the tuning forks say, A is struck on a rubber pad, it starts vibrating. On putting the tuning fork A on its sound box, we find that the other tuning fork B also starts vibrating and a loud sound is heard.**



Reason: The vibrating tuning fork A produces the forced vibrations in the air column of its sound box. These vibrations are of large amplitude because of large surface area of air in the sound box. They are communicated to the sound box of the fork B. The air column of B starts vibrating with the frequency of fork A. Since the frequency of these vibrations is same as the natural frequency of the fork B, the fork B picks up these vibrations and starts vibrating under resonance. Thus the two sound boxes help in communicating the vibrations and in increasing the amplitude of vibrations.

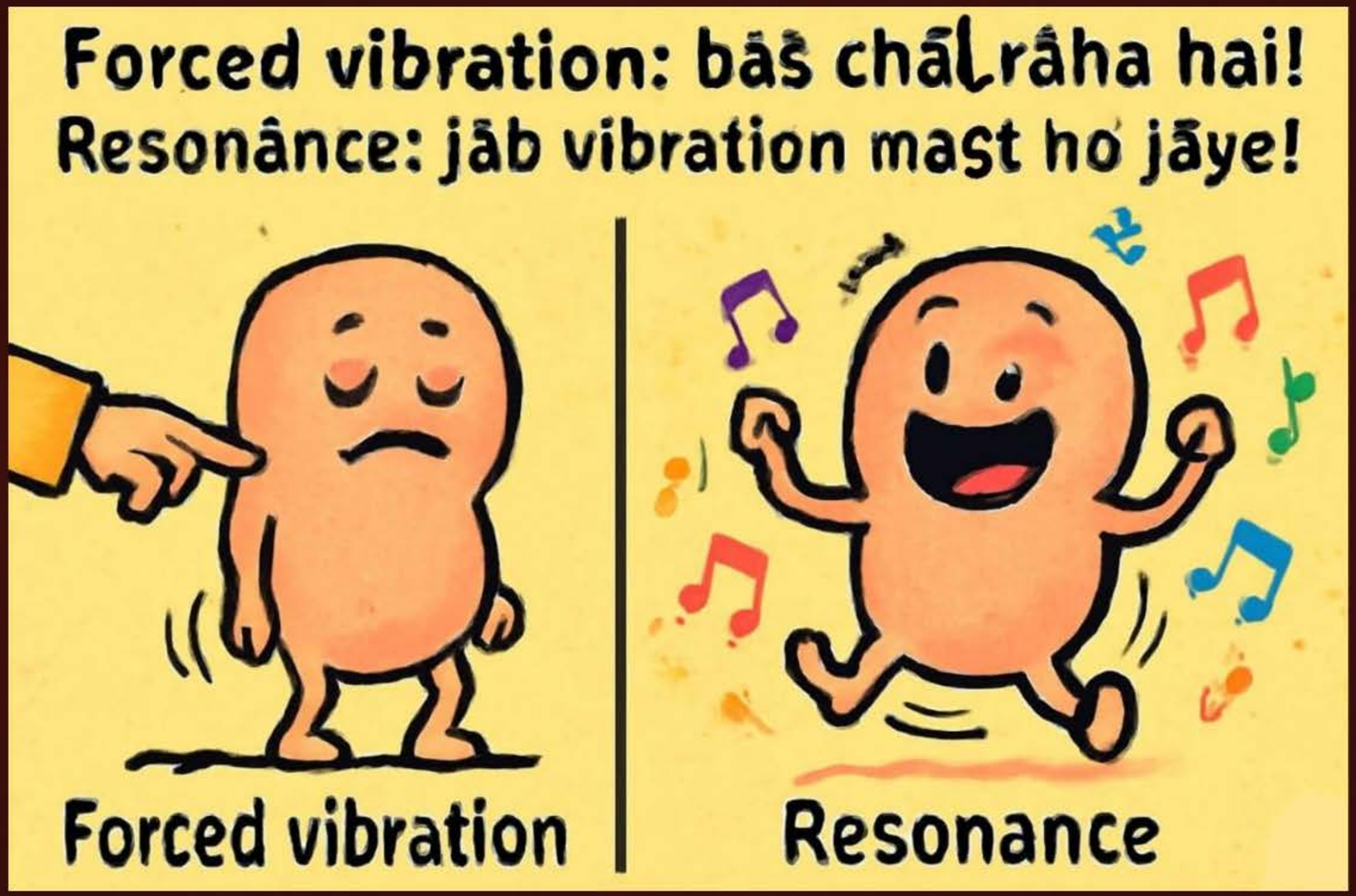

RESONANCE:
jab mere vibes
tumhare vibes se mil jayein!

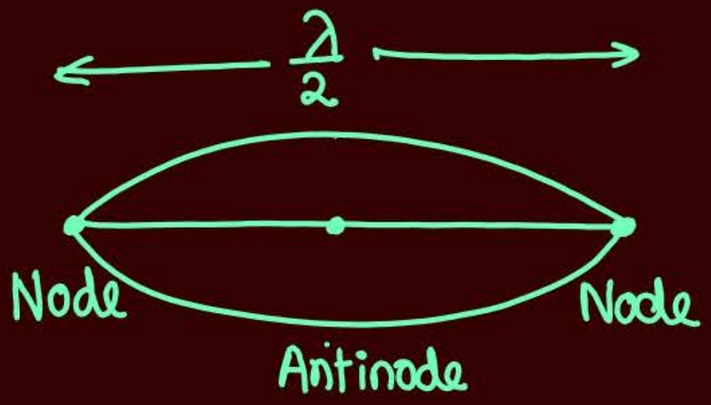


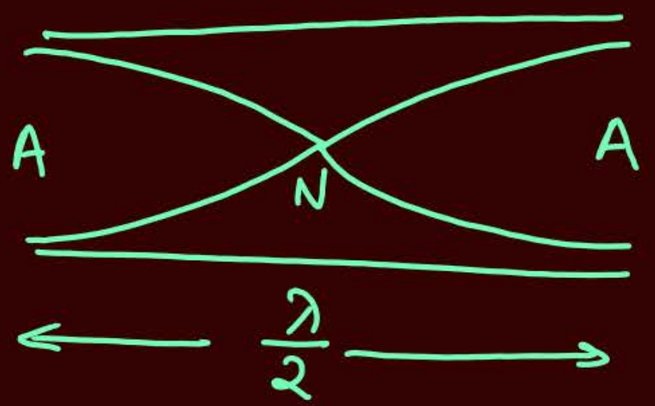
Difference between the Forced and Resonant Vibrations

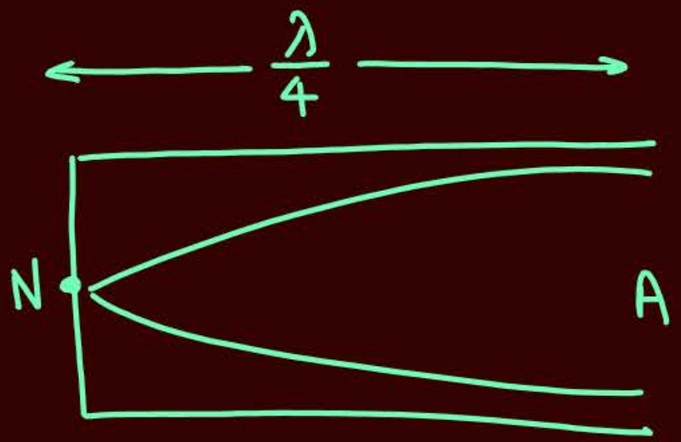
HW

Forced vibrations	Resonant vibrations
1. The vibrations of a body under an external periodic force of frequency different from the natural frequency of the body, are called the forced vibrations.	1. The vibrations of a body under an external periodic force of frequency exactly equal to the natural frequency of the body, are called the resonant vibrations.
2. The amplitude of vibration is usually small.	2. The amplitude of vibration is very large.
3. The vibrations of the body are not in phase with the external periodic force.	3. The vibrations of the body are in phase with the external periodic force.
4. These vibrations last for a very small time after the periodic force has ceased to act.	4. These vibrations last for a long time after the periodic force has ceased to act.


Forced vibration: bās chāl rāha hai!
Resonance: jāb vibration mast ho jāye!
Forced vibration
Resonance







$V = f\lambda$

$$f = \frac{V}{\lambda}$$

QUESTIONS

Fig. shows three different modes of vibration of a string of length *l*.

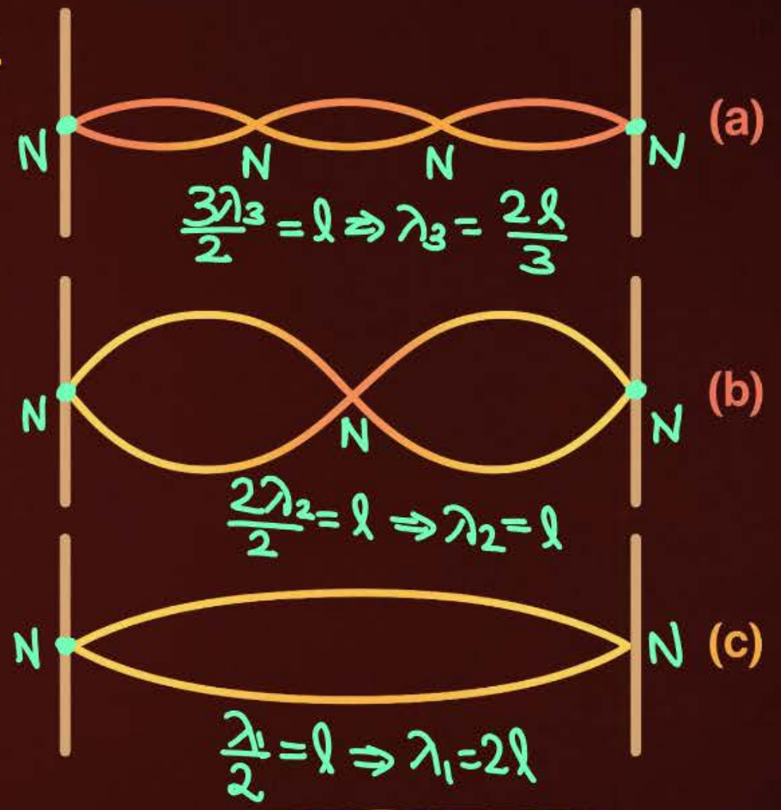
(i) Which of the vibration is of largest amplitude? (B)

(ii) Which of the vibration is of least frequency? (C)

(iii) What is the ratio of frequency between (a) and (c)? 3:1

(iv) What is the ratio of wavelength between (b) and (a)? 3:2

$$\frac{\lambda_2}{\lambda_3} = \frac{l}{\frac{2l}{3}} = \frac{3}{2}$$

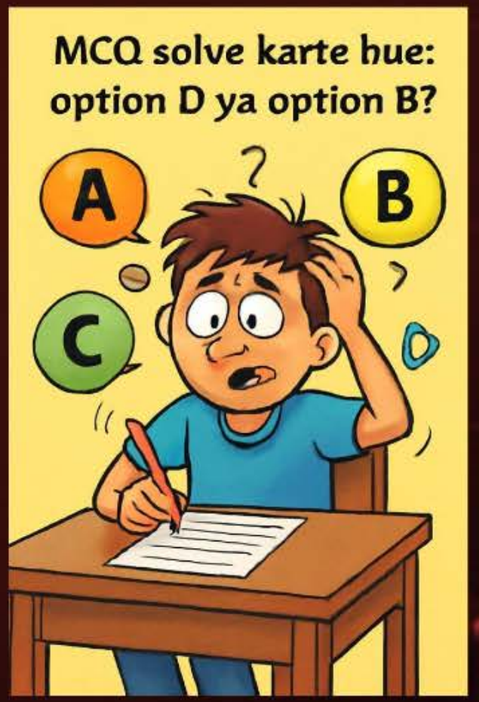




QUESTIONS

A wire stretched between the two fixed supports, is plucked exactly in the middle and then released. It executes (neglect the resistance of the medium):

1. resonant vibrations
2. natural vibrations
3. damped vibrations
4. forced vibrations





QUESTIONS

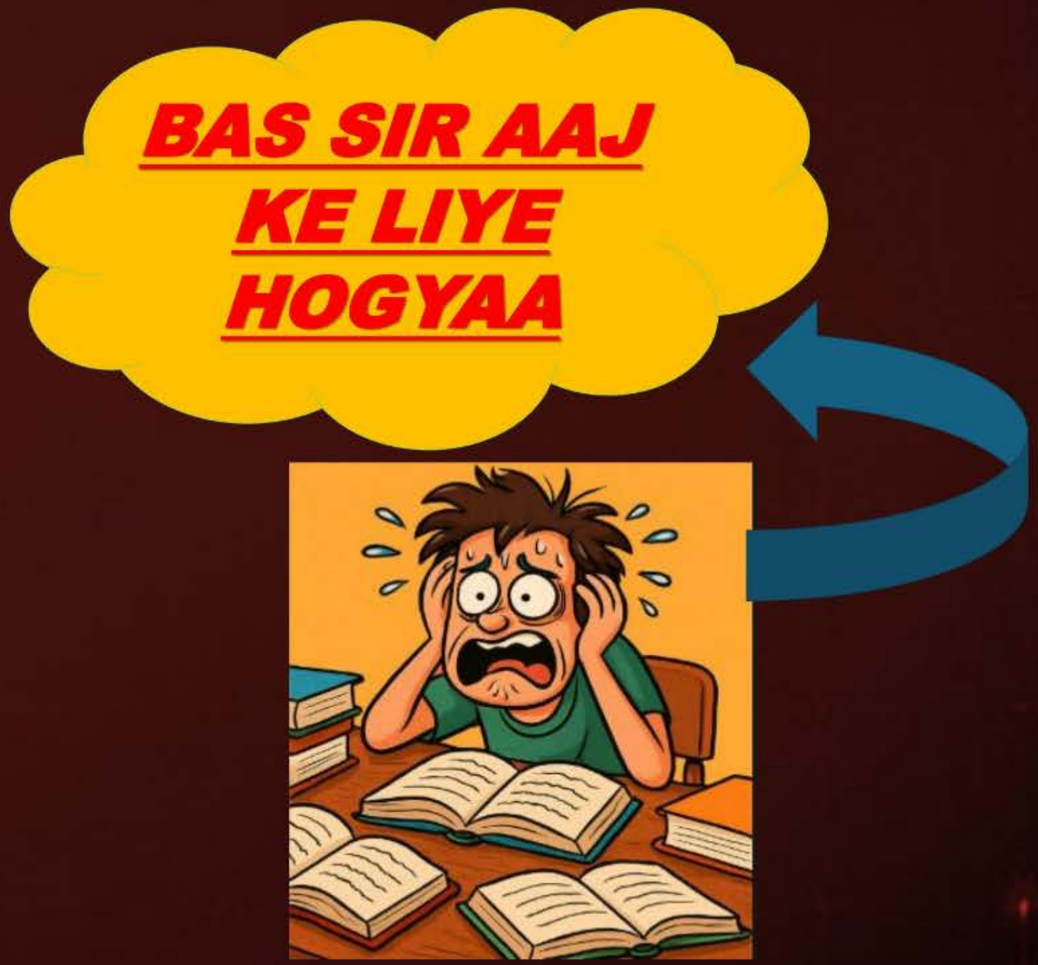
When a body vibrates under a periodic force, the vibrations of the body are:

1. natural vibrations
2. damped vibrations
3. forced vibrations
4. resonant vibrations

QUESTIONS

A tuning fork of frequency 256 Hz will resonate with another tuning fork of frequency:

1. 128 Hz
2. 256 Hz
3. 384 Hz
4. 512 Hz



HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

QUESTIONS

The stem of a vibrating tuning fork is pressed against a table top.

(a) Will it produce an audible sound?

(b) Does it cause the table top to set in vibrations? If yes, what type of vibrations are they?

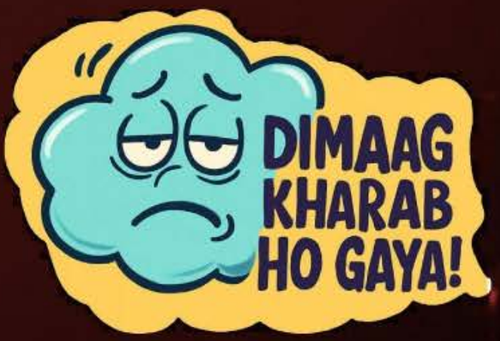
(c) Under what condition does it lead to resonance?

Solution:

(a) Yes.

(b) Yes, the vibrations caused in the table top are the forced vibrations.

(c) It leads to resonance if the natural frequency of vibrations of the table top becomes equal to the frequency of the vibrating tuning fork.



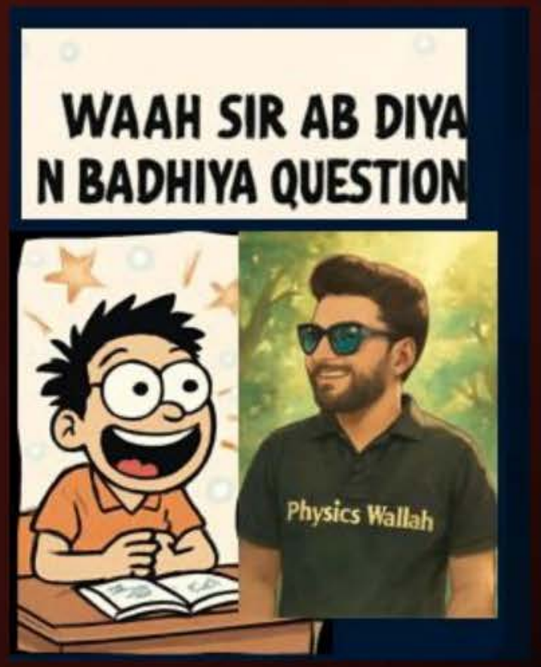
QUESTIONS

Sometimes when a vehicle is driven at a particular speed, a rattling sound is heard. Explain briefly, why does this happen and give the name of the phenomenon taking place. Suggest one way by which the rattling sound could be stopped.

Solution:

At a particular speed of the vehicle, when the natural frequency of vibration of some parts of the vehicle becomes equal to the frequency of the to and fro movement of the piston of its engine, resonance occurs due to which those parts of the vehicle start vibrating violently and a rattling sound is heard.

To stop the rattling sound, the speed of the vehicle should be changed.



Thank You